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% GENERATED BY BING COPILOT (FREE)
% Provided my red_svd() function to CoPilot and prompted it to create a
% function that plots the average values across all datasets for MSE, PSNR,
% and file size for a vector of k values. I then did a follow up prompt for
% it to plot the average values for each dataset separately on the same
% graph.
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function plot_results(k_values, num_samples)
    % plot_svd_tradeoff: Evaluate SVD compression trade-offs
    % k_values: vector of ranks to test
    % num_samples: number of random images to sample per dataset

    rng(441); % reproducibility
    dataset_paths = { 'landscape/color/', 'butterfly/', 'baseball/' };
    dataset_names = { 'Landscape', 'Butterfly', 'Baseball' };

    % Storage for results
    mse_all = zeros(length(k_values), length(dataset_paths));
    psnr_all = zeros(length(k_values), length(dataset_paths));
    size_all = zeros(length(k_values), length(dataset_paths));

    % Loop over datasets
    for d = 1:length(dataset_paths)
        folder = dataset_paths{d};
        files = dir(fullfile(folder, '*.jpg'));
        num_files = length(files);
        rand_idx = randperm(num_files, min(num_samples, num_files));

        mse_vals = zeros(length(k_values), length(rand_idx));
        psnr_vals = zeros(length(k_values), length(rand_idx));
        size_vals = zeros(length(k_values), length(rand_idx));

        % Loop over sampled images
        for f = 1:length(rand_idx)
            filename = fullfile(folder, files(rand_idx(f)).name);
            orig = imread(filename);
            Im = double(orig);

            % Loop over ranks
            for r = 1:length(k_values)
                k = k_values(r);
                Imapprox = zeros(size(Im));
                for i = 1:size(Im,3)
                    [U,S,V] = svd(Im(:, :, i));
                    Imapprox(:, :, i) = U(:, 1:k)*S(1:k, 1:k)*V(:, 1:k)';
                end
                approx_uint8 = uint8(Imapprox);

                % Save temporarily to measure file size
                outFile = [tempname '.png'];
                imwrite(approx_uint8, outFile);
                fileInfo = dir(outFile);
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        fileSizeKB = fileInfo.bytes / 1024;
        delete(outFile);

        % Compute metrics
        mse_vals(r,f) = immse(approx_uint8, orig);
        psnr_vals(r,f) = psnr(approx_uint8, orig);
        size_vals(r,f) = fileSizeKB;
    end
end

% Average across sampled images
mse_all(:,d) = mean(mse_vals,2);
psnr_all(:,d) = mean(psnr_vals,2);
size_all(:,d) = mean(size_vals,2);
end

% Aggregate across datasets
mse_mean = mean(mse_all,2);
psnr_mean = mean(psnr_all,2);
size_mean = mean(size_all,2);

% == Aggregate plots ==
figure;
plot(k_values, mse_mean, '-o'); grid on;
xlabel('Rank k'); ylabel('MSE');
title('Average MSE vs Rank (All Datasets)');

figure;
plot(k_values, psnr_mean, '-o'); grid on;
xlabel('Rank k'); ylabel('PSNR (dB)');
title('Average PSNR vs Rank (All Datasets)');

figure;
plot(k_values, size_mean, '-o'); grid on;
xlabel('Rank k'); ylabel('File Size (KB)');
title('Average File Size vs Rank (All Datasets)');

% == Per-dataset plots ==
figure;
hold on;
for d = 1:length(dataset_paths)
    plot(k_values, mse_all(:,d), '-o', 'DisplayName', dataset_names{d});
end
grid on; xlabel('Rank k'); ylabel('MSE');
title('Average MSE vs Rank (Per Dataset)');
legend show; hold off;

figure;
hold on;
for d = 1:length(dataset_paths)
    plot(k_values, psnr_all(:,d), '-o', 'DisplayName', dataset_names{d});
end
grid on; xlabel('Rank k'); ylabel('PSNR (dB)');
title('Average PSNR vs Rank (Per Dataset)');

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    legend show; hold off;

    figure;
    hold on;
    for d = 1:length(dataset_paths)
        plot(k_values, size_all(:,d), '-o', 'DisplayName', dataset_names{d});
    end
    grid on; xlabel('Rank k'); ylabel('File Size (KB)');
    title('Average File Size vs Rank (Per Dataset)');
    legend show; hold off;
end
```

Not enough input arguments.

Error in plot_results (line 18)
 mse_all = zeros(length(k_values), length(dataset_paths));
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